An exact Mumford count differs from one half

For $f = X^5 - X$ over $\mathbf{F}_3$, Lean counts the complete space of polynomial pairs with u monic, $\deg u \leq 2$, $\deg v < \deg u$ and $u \mid v^2 - f$. There are eight such reduced pairs; seven have completely split u. Identity $(1, 0)$ is included. [pcc:unverified]

$$\text{split fraction} = \frac{7}{8}, \qquad \frac{7}{8} - \frac{1}{2} = \frac{3}{8}.$$

This disproves an exact finite-field reading of $1/g!$ at genus two. Gaudry's fixed-genus, $q \to \infty$ limit is preserved. The counted object is the complete reduced-Mumford polynomial-pair model; formal geometric identification with $\mathrm{Pic}^0(C)$ remains open. [pcc:unverified]

The curve polynomial is proved monic, degree five and separable/squarefree. An explicit bijection proves that every admissible polynomial pair has a canonical coefficient shape and that distinct shapes give distinct pairs. The enumeration does not assume a selected list is complete. [pcc:unverified]

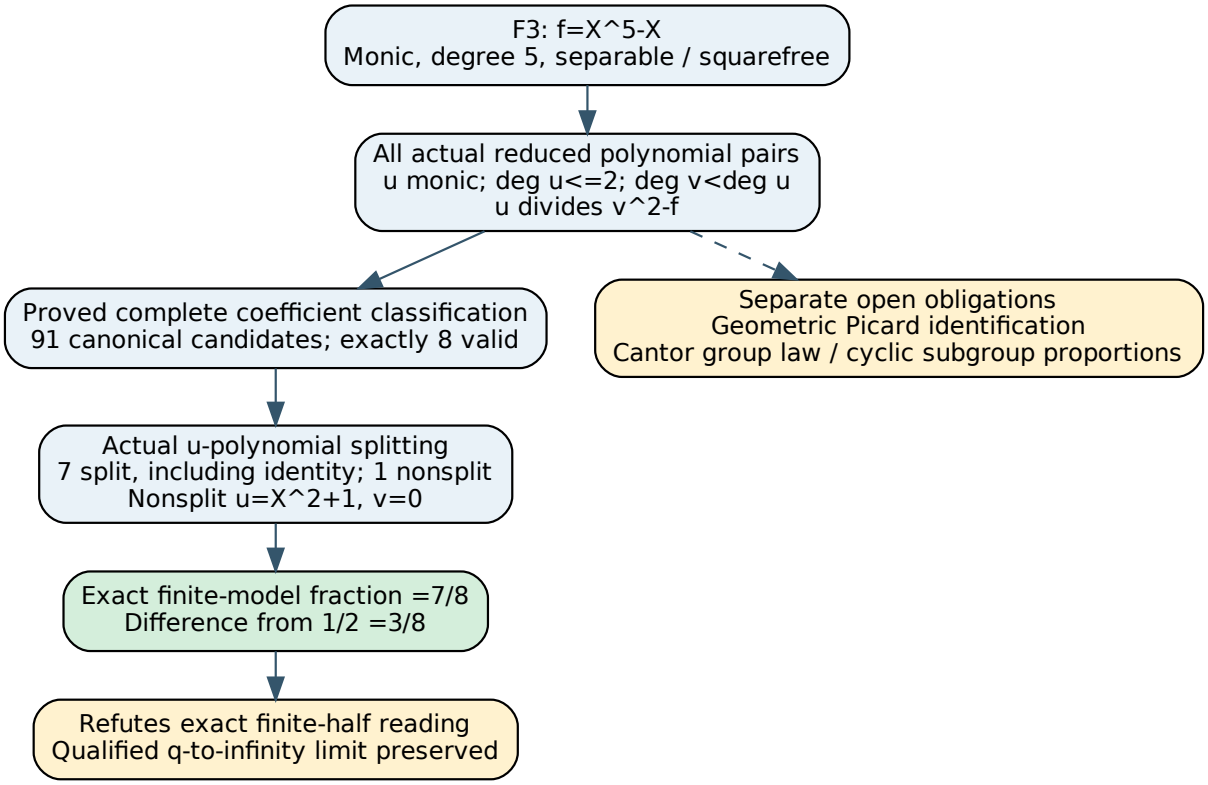


Figure 1: Complete pair model, exact splitting count and separate open obligations.

Proof mechanics, scope and checks

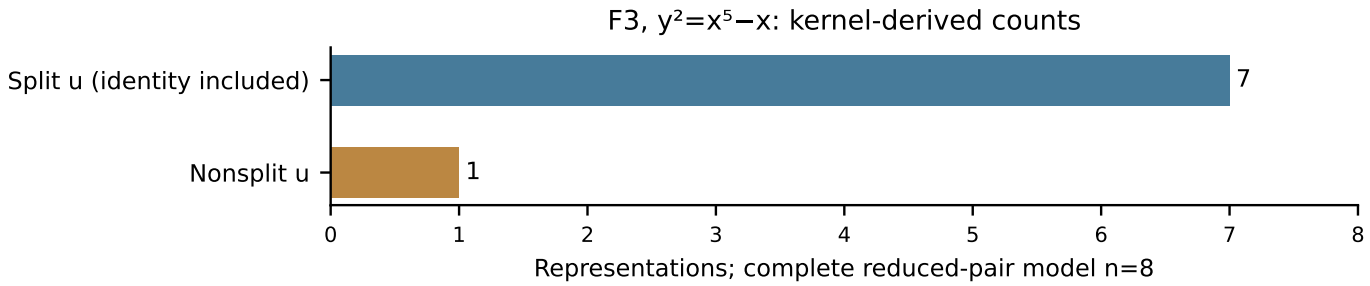


Figure 2: Exact proof counts; all eight representations, without statistical sampling.

Of 91 canonical coefficient candidates, eight satisfy the actual reduced pair conditions. Seven have split u; the sole nonsplit case is u=X²+1,v=0. [pcc:unverified]

Mathlib's polynomial division is noncomputable. Lean first proves a scalar coefficient test equivalent to actual monic division, using an exact quotient/remainder identity and a degree bound. Kernel `decide` enumerates the scalar test, and proved equivalences transfer its count to the full polynomial-pair space. Splitting is the actual `Polynomial.Splits` predicate: $X^2 + 1$ has no root over F3, and the other seven cases split. [pcc:unverified]

Source: ../../lean/MumfordSmoothness.lean. The combined binding is `MumfordSmoothnessAudit.mumford_count_audit_certificate`. The whole tracked project gate passes 80 declarations, 0 failing and 0 trusted-native exceptions; counts include generated declarations. Only propext, Classical.choice and Quot.sound are accepted. [pcc:unverified]

Source commit 6fee3178 precedes issue, including the standalone issuer's dependency lock. Packet PCC-20261109-07eb0d pins the combined theorem; its full local check passes20:00:58Z. The receipt is ../../evidence/pcc-mumford-count-check.json. Standing remains unsigned until a separate signed verdict. [pcc:unverified]

Supplement IDC1h7d12c937b3b1482b checks only the scalar quotient/remainder rewrite: 5 variables, total degree bound 5, 32 derived and 11 extra points over $2^{61}-1$. The unchanged Rust checker accepts, and the source pointer matches. These finite one-host samples establish neither F3 cardinalities nor splitting. The pair counts and splitting rest on Lean; geometry and the subgroup law remain open. [pcc:unverified]

Primary read scopes: Gaudry2000 Definitions 2–4, Section 2.3 and Proposition 4; MMT2002 Sections 4.2–4.3, printed pp133–134/PDF7–8. The MMT PDF page8 was visually inspected: eq12 begins at degree1 and omits identity; equality of subgroup and full-group proportions is explicitly a heuristic before eq13. This odd-characteristic F3 count does not test MMT's binary large-prime regime. URLs and exact hashes are in ../../sources.json. [pcc:unverified]

Cantor arithmetic, cyclic-subgroup proportions, geometric Picard identification, a scaling family and a native multi-field census remain open. No new native population, statistical study, benchmark, paid compute or separate-host run occurred; the original fourteen-item request remains partial. [pcc:unverified]

All nine attempt logs remain: four passing stages/five failed builds, then the passing gate. No performance or curve-comparison metric changed. [pcc:unverified]